



GUIDELINE

Agitation and Arousal Medication Management

Scope (Staff):	Medical, Nursing, Pharmacy
Scope (Area):	All clinical areas

Child Safe Organisation Statement of Commitment

CAHS commits to being a child safe organisation by applying the National Principles for Child Safe Organisations. This is a commitment to a strong culture supported by robust policies and procedures to reduce the likelihood of harm to children and young people.

This document should be read in conjunction with this [disclaimer](#)
If you have any feedback or comments to improve our clinical policies or our systems, please tell us by filling in the [feedback form](#).

Contents

Aim	2
General Principles	2
Risk	2
Key Points.....	2
Documentation.....	2-3
Pharmaceutical Management Algorithm.....	4
Medication Choices for Patients 12 Years and Older	5-6
Medication Choices for Patients 6-11 Years Old	7-8
Monitoring for Adverse Effects	9-10
Post Sedation Care.....	10
Post Sedation Care.....	10-11
Reversal Agents for Overdose	11-12
Additional Monitoring	12
Follow Up	12

Aim

This guideline is for the stepwise management of acute arousal and agitation in children and adolescents, when non-drug approaches are unsuccessful. Doses need adjusting according to weight, age, and co-morbid physical illness.

General Principles

Other interventions (i.e. behavioural), need to be used as part of the overall management of the acutely aroused and agitated patient, together with the use of medication as an adjunct to this.

This paediatric clinical practice guideline has been developed to assist clinicians with decisions regarding appropriate medication interventions to treat agitation and arousal. It does not indicate an exclusive course of action or standard of care. It does not replace the need for application of clinical judgement to each individual presentation.

Risk

Failure to adhere to this guideline may result in harm to patients and staff or damage to property due to insufficient or excessive use of medication in the management of paediatric arousal and agitation.

Key Points

- This guideline provides information on the drugs, doses, monitoring and treatment of adverse events in the management of arousal and agitation in children and adolescents at CAHS.
- This guideline is only of relevance in environments where clinical expertise and technical equipment is available to ensure the safe administration and monitoring of patients administered medications as per this guideline. For all other areas follow your local policy.

Documentation

Medications are to be prescribed on the “PRN” side of the paediatric National Inpatient Medication Chart (pNIMC). The prescriber should annotate “Step 1”, “Step 2”, “Step 3” or “Step 4” as appropriate in the “Pharmacy / Additional Information” section of medication order. Alternatively, the Agitation and Arousal Medication Chart can be used in areas where its use is approved (e.g. Ward 4A, Ward 5A and Clinic K).

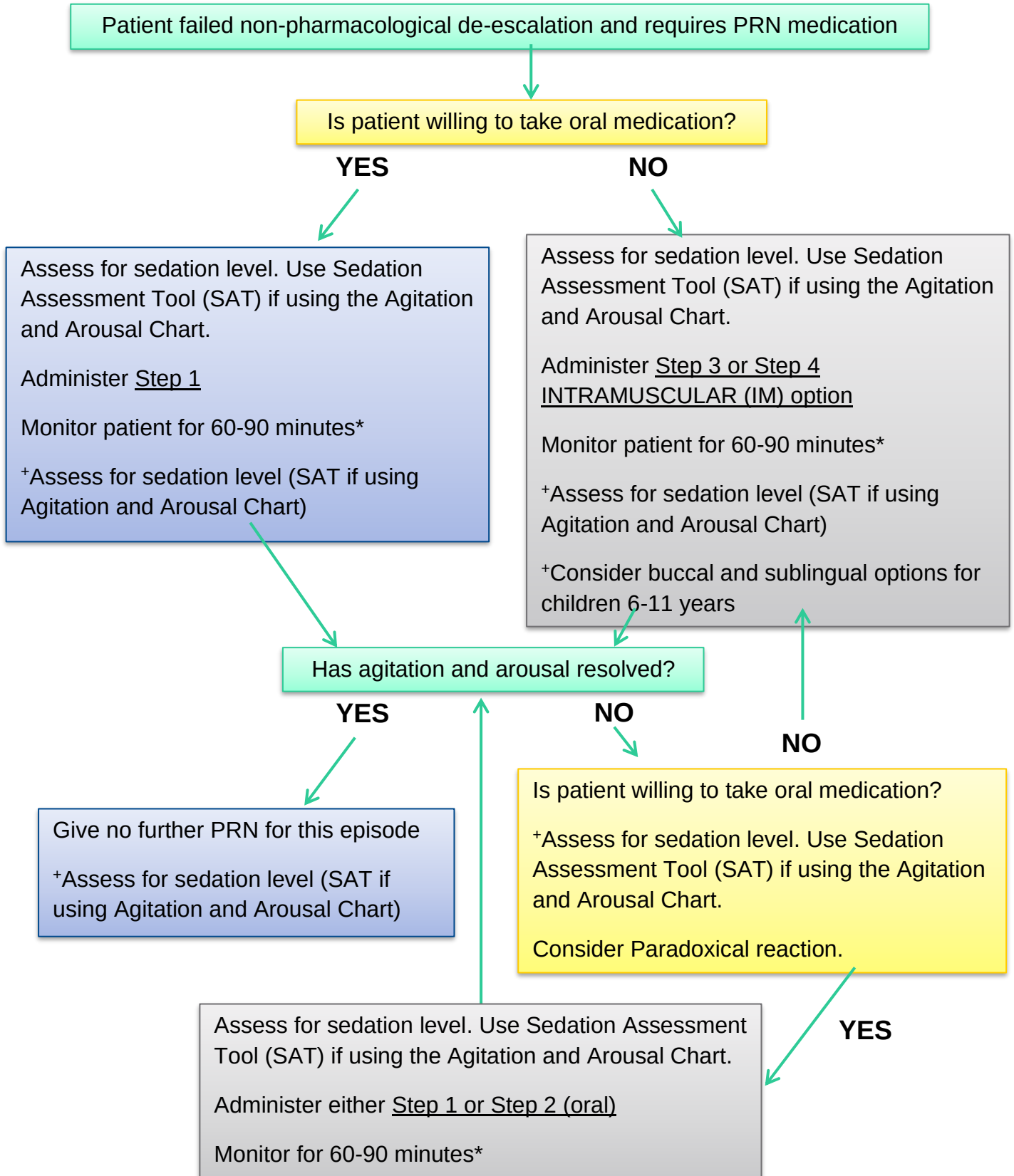
Example: Agitation Medications prescribed on "PRN" section of pNIMC

AFFIX PATIENT IDENTIFICATION LABEL HERE AND OVER LEAF				Attach ADR Sticker See front page for details			
Med Rec. No: 217865				AS REQUIRED "PRN" MEDICATIONS			
Surname: FROG							
Forename: KERMIT							
Gender: MALE D.O.B. 01/08/2011							
1st Prescriber to print patient name & check label correct: <i>Kermit Frog</i>							
YEAR 20 21				WARD/UNIT: 1B			
				Weight (kg) 35			
				Date 17/11/2021			

VALID ORDER UNLESS LEGIBLE	Date: 17/11 Medication (Print Generic Name): CLONIDINE	Date:	BINDING MARGIN - DO NOT WRITE		
	Route: PO DOSE: 50 MICROGRAMS Hourly Frequency: 4 HRLY PRN: 200 MICROGRAM Max DOSE/24hrs:	Time:		Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No	
	Pharmacy/Additional Information: STEP 1 (Not within 2 hours of regular dose) VM 17/11			DOSE:	
	Indication: Agitation and Arousal Calculation of Dose (mg/kg/dose): 1.4 micrograms/kg/dose			Route:	
	Prescriber Signature: <i>A DOC</i> Print Name: A DOC Contact/Pager: 56897	Sign:		Continue on discharge? Dispense? Duration:	
	Date: 17/11 Medication (Print Generic Name): LORAZEPAM	Date:		Pharmacist Date Contact number	
	Route: PO DOSE: 1 MG Hourly Frequency: 2 HRLY PRN: 4 MG Max DOSE/24hrs:	Time:			Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No
	Pharmacy/Additional Information: STEP 2 (Not within 2 hours of regular dose) VM 17/11				DOSE:
	Indication: Agitation and Arousal Calculation of Dose (mg/kg/dose): 28 micrograms/kg/dose				Route:
	Prescriber Signature: <i>A DOC</i> Print Name: A DOC Contact/Pager: 56897	Sign:			Continue on discharge? Dispense? Duration:
Date: 17/11 Medication (Print Generic Name): OLANZAPINE	Date:	Pharmacist Date Contact number			
Route: IM DOSE: 2.5 MG Hourly Frequency: 6 HRLY PRN: 5 MG Max DOSE/24hrs:	Time:		Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No		
Pharmacy/Additional Information: STEP 3 (Not within 1 hour of IM midazolam) VM 17/11			DOSE:		
Indication: Agitation and Arousal Calculation of Dose (mg/kg/dose):			Route:		
Prescriber Signature: <i>A Doc</i> Print Name: A Doc Contact/Pager: 56897	Sign:		Continue on discharge? Dispense? Duration:		
Date: 17/11 Medication (Print Generic Name): MIDAZOLAM	Date:		Pharmacist Date Contact number		
Route: IM DOSE: 2.5 MG Hourly Frequency: 2 HRLY PRN: 5 MG Max DOSE/24hrs:	Time:			Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No Yes / No	
Pharmacy/Additional Information: STEP 4 (Not within 1 hour of IM olanzapine) VM 17/11				DOSE:	
Indication: Agitation and Arousal Calculation of Dose (mg/kg/dose): 70 micrograms/kg/dose				Route:	
Prescriber Signature: <i>A DOC</i> Print Name: A DOC Contact/Pager: 56897	Sign:			Continue on discharge? Dispense? Duration:	

Pharmaceutical Management Algorithm for a new episode of Agitation and Arousal

To be used in conjunction with [Arousal and Agitation Management for Mental Health Inpatients Guideline](#).



Medication Choices for Patients 12 Years and Older

Step 1 - For patients able and agreeable to take ORAL medication	
<i>Short acting oral benzodiazepines such as lorazepam should be considered as first line agents, particularly for antipsychotic naïve patients.</i> Use Ideal Body Weight for benzodiazepines. Refer to Guidelines for Drug Dosing in Overweight and Obese Children 2 to 18 Years of Age.	
Oral Lorazepam	0.5 - 1mg PO initially; (up to 2mg as a single dose in highly agitated patients). Do not exceed 50 micrograms/kg/dose.¹⁻³ Maximum of 4mg/24 hours.¹⁻³
Step 2 - If greater anxiolytic / sedative effects are required and the patient <u>is</u> agreeable to take oral medication ⁸ :	
Oral Olanzapine (wafers recommended over tablets)	<40kg: 2.5 - 5mg PO.^{1, 4} - Maximum of 10mg/24 hours.^{1, 4} >40kg: 5 - 10mg PO.^{1, 4} - Maximum of 20mg/24 hours.^{1, 4}
<u>OR</u>	
Oral Quetiapine	12.5mg – 25mg PO initially; (up to 50mg per dose in highly agitated patients)^{4, 5} Maximum of 200mg/ 24 hours.^{4, 5}
Review after 30 - 60 minutes, repeat dose given at Step 2 if necessary. <i>Note – these may be similar sedative doses but should not be considered as having equivalent antipsychotic effects.</i>	
Step 3 & 4 - If greater anxiolytic / sedative effects required <u>and</u> patient not agreeable to oral medication ⁸ :	
Consider IM benzodiazepine as first line.	
Intramuscular Midazolam	2.5mg – 5mg IM (50 to 150 microgram/kg to a max of 10mg per dose in highly agitated patients).^{1, 2, 6} < 16 years: Maximum of 10mg/24 hours.^{1, 2, 6} ≥ 16 years: Maximum of 20mg/24 hours.^{1, 2, 6}
<u>OR</u>	

Intramuscular Olanzapine	2.5mg – 5mg IM (up to 10mg per dose in highly agitated patients).^{1,4} • < 16 years: Maximum of 10mg/24 hours.^{1, 4} • ≥ 16yrs: Maximum of 20mg/24 hours.^{1, 4} <i>Higher doses can be used <u>upon specialist advice</u> in highly distressed patients (e.g. eating disorder patients requiring sedation to accept nasogastric feeds).²</i> <i>> 12 years of age and > 40kg; up to 30mg/24 hours. Do not give a second (10mg) dose more frequently than 2 hours after the initial dose or 4 hours after the second dose.²</i> Olanzapine reconstitution One vial reconstituted with 2.1mL of Water for Injections gives a 10mg/2mL solution.⁷ <i>Dose of Olanzapine may be increased at intervals of 2.5mg-5mg.</i>^{1, 4}
<u>OR</u>	
Intramuscular Clonazepam	1 - 2mg IM per dose. • ≤ 12 years: Maximum of 2mg/24 hours.⁷ • > 12 years: Maximum of 4mg/24 hours.⁷
<i>Review after 30 minutes, repeat dose given at Step 3 if necessary.</i> <i>Note – clonazepam and olanzapine have a long half-life and caution should be taken to avoid multiple injections within a short time frame due to risk of accumulation.</i> <u>Do not exceed maximum 24 hour dosages.</u>²	
<i>Do not administer both IM olanzapine and IM benzodiazepine together.</i> <i>If the use of both is required, they must be separated by at least one hour. Monitor for over-sedation, respiratory and cardiovascular depression using close physical observations and Observation and Response Chart as detailed in the monitoring section below.⁸</i>	

Note: This paediatric clinical practice guideline has been developed to assist clinicians with decisions regarding appropriate medication interventions to treat agitation and arousal. It does not indicate an exclusive course of action or standard of care. It does not replace the need for application of clinical judgement to each individual presentation.

Medication Choices for Patients 6-11 Years Old

Step 1 - For patients able and agreeable to take ORAL medication	
Consider an α_2 agonist such as clonidine as a first line agent. If the patient is already taking clonidine or guanfacine a short acting oral benzodiazepine such as lorazepam should be considered. Use Ideal Body Weight for benzodiazepines. Refer to Guidelines for Drug Dosing in Overweight and Obese Children 2 to 18 Years of Age	
Oral Clonidine	1 - 2 micrograms/kg PO initially (up to 4 micrograms/kg [max 150 micrograms] per dose in highly agitated patients).¹⁻³ 20 – 40kg: Maximum of 200 micrograms/24 hours.² 40 – 45kg: Maximum of 300 micrograms/24 hours.² > 45 kg: Maximum of 400 micrograms/24 hours.²
OR	
Oral Lorazepam	0.5 – 1mg PO initially; (up to 2mg as a single dose in highly agitated patients). Do not exceed 50 micrograms/kg/dose.¹⁻³ Maximum of 4mg/24 hours.¹⁻³
Step 2 - If greater anxiolytic / sedative effects are required and the patient <u>is</u> agreeable to take oral medication:	
Oral Quetiapine	12.5mg – 25mg PO initially; (up to 50mg per dose in highly agitated patients)^{4, 5} Maximum of 200mg/24 hours.^{4, 5} <i>*Higher doses can be used under consultant supervision*</i>
Review after 30 - 60 minutes, repeat dose given at Step 2 if necessary. <i>Note – blood pressure should be monitored if there is co-administration of clonidine in Step 1 and quetiapine in Step 2 as these can have additive agonistic effects on α_2 receptors.</i>	
Step 3 - If greater anxiolytic / sedative effects required <u>and</u> patient not agreeable to oral medication:	
Consider IM benzodiazepine as first line.	
Intramuscular/Buccal Midazolam	2.5mg – 5mg buccal/IM (50 to 150 microgram/kg to a max of 10mg per dose in highly agitated patients).^{1, 2, 6} Maximum 10mg/24 hours.^{1, 2, 6}

Step 4 - If greater anxiolytic / sedative effects required <u>and</u> intramuscular midazolam has been ineffective:	
Oral/Intramuscular/ Sublingual Olanzapine	2.5mg – 5mg PO/IM/Sublingual (up to 10mg per dose in highly agitated patients).^{1, 4} Maximum of 10mg/24 hours.^{1, 4} Olanzapine reconstitution One vial reconstituted with 2.1mL of Water for Injections gives a 10mg/2mL solution).⁹ <i>Dose of Olanzapine may be increased at intervals of 2.5mg-5mg.⁴</i>
<i>Review after 30 minutes, repeat dose given at Step 3 if necessary.</i> <i>Note – Olanzapine has a long half-life and caution should be taken to avoid multiple injections within a short time frame due to risk of accumulation.²</i> <u>Do not exceed maximum 24 hour dosages.</u>	
<i>Do not administer both IM olanzapine and IM benzodiazepine together.</i> <i>If the use of both is required, they must be separated by at least one hour. Monitor for over-sedation, respiratory and cardiovascular depression using close physical observations and Observation and Response Chart as detailed in the monitoring section below.⁸</i>	
Step 5 - Ensure maximum 24 hour dosages have been given in Steps 1, 2, 3 & 4.	
Step 6 - If agitation has not resolved contact treating consultant/psychiatrist and/or anaesthetist.	

Note: This paediatric clinical practice guideline has been developed to assist clinicians with decisions regarding appropriate medication interventions to treat agitation and arousal. It does not indicate an exclusive course of action or standard of care. It does not replace the need for application of clinical judgement to each individual presentation.

Monitoring: For Adverse Effects

Antipsychotics

Children and adolescents are at greater risk for acute dystonic reactions. Children may not complain of akathisia in the same way as adults therefore close monitoring is recommended.

Monitor for:

- Excessive sedation
- Respiratory depression

- Airway obstruction
- Postural/Orthostatic Hypotension
- Extrapyramidal side effects
 - Akinesia: Finding it hard to start a movement
 - Akathisia: Finding it hard to keep still, with a feeling of restlessness
 - Dyskinesia: Usual movements or twitches (usually of the face) that may be repetitive
 - Oculogyric crisis: Unusual eye movements, most commonly with the eyes turning upwards
 - Parkinsonism: Symptoms look like someone with Parkinson's Disease e.g. tremor, stiffness
 - Dystonia: Muscle stiffening
- [Neuroleptic malignant syndrome](#)
- Anticholinergic side effects (blurred vision, mydriasis, constipation, nausea, dry mouth, urinary retention).
- Tachycardia with the potential for QT prolongation

Benzodiazepines

Monitor for:

- Excessive sedation
- Respiratory depression
- Airway obstruction
- Disinhibition and delirium
- Hypotension
- Hypersalivation
- Blurred vision
- Loss of coordination and memory loss
- Paradoxical reaction (agitation/hyperactivity)

Alpha₂ Agonists

Monitor for:

- Excessive sedation
- Hypotension

- Bradycardia
- Fluid retention
- Airway obstruction

Post Sedation Care

If decreased level of consciousness:

Commence 1:1 nursing observations of patient and document on the Observation and Response Chart and escalate to medical referral as appropriate:

1. Check:
 - a. Conscious state.
 - b. Vital signs.
 - c. Cardiac monitoring.
 - d. Respiratory function.
 - e. Signs of upper airway obstruction and
 - f. Extrapramidal effects.
2. Support airway and supplement oxygen as per hospital protocol with continuous oxygen saturation monitoring.
3. Place the patient in recovery position.
4. Perform physical examination.
5. Take collateral history.

Sedated patients should be monitored in an appropriate clinical area with resuscitation facilities.

Suggested frequency

- Initial continuous observation during first 10 minutes.
- Then every 10 minutes for 30 min.
- Then every 15 min for 30 min.
- Then every 30 min for 60 min.
- Then hourly for 4 hours or until awake.

Consider

- *Head injury.*
- *Metabolic causes.*
- *Substance abuse.*

Reversal Agents for Overdose

Treatment for adverse events	Dose and Frequency
Benzodiazepine Overdose	<u>For Children 1-18 years:</u> • Call 55 Code Blue Medical if appropriate. • Use intravenous Flumazenil.¹⁰ • 5-10 microgram/kg (maximum 200 micrograms) over 15 seconds.⁴ • If response is inadequate, repeat dose every 60 seconds to a total of 50 microgram/kg up to 1 mg.⁴ • If required, follow with an IV infusion of 2 to 10 micrograms/kg/hour, titrated to response (maximum of 400 micrograms/hour).⁴
Extrapyramidal Side Effects (EPSE)	<u>For Children 3 -11 years:</u> • Call 55 Code Blue Medical if appropriate. • Use Benzatropine.⁴ • Oral: 20 - 50 microgram/kg (maximum 2mg) every 12 to 24 hours. Round dose to the nearest 500 microgram.⁴ OR • Intravenous or Intramuscular: 20 microgram/kg (maximum 1mg) – may be repeated after 15 minutes.⁴ <i>Maximum total oral / IM / IV dose is 4mg/24 hours.⁴</i> <u>For Children ≥ 12-18 years:</u> • Call 55 Code Blue Medical if appropriate. • Use Benzatropine.⁴ • Oral: 1 - 2 mg/dose, repeated every 12 - 24 hours.⁴ OR • Intravenous or Intramuscular: 1 - 2 mg/dose - may be repeated after 15 minutes.⁴ <i>Maximum total oral / IM / IV dose is 4mg/24 hours.⁴</i>

Additional Monitoring

Refer to [Monitoring the effects of psychotropic medication on physical health](#)

Follow Up

- The Chief Psychiatrist recommends that there is regular screening for metabolic syndrome risk factors in all patients prescribed antipsychotic medications regularly and that this is documented in the patient medical record including:
 - Monitoring of blood glucose and serum lipids;
 - Monitoring and management of weight gain, including attention to weight loss or prevention of weight gain, waist circumference monitoring; and
 - Regular measurement and monitoring of blood pressure.
- All patients who are discharged on regular antipsychotic medication, should be followed up in accordance to the above recommendations and current policies.

References and related external legislation, policies, and guidelines

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4. Australian Medicines Handbook Children's Dosing Companion Adelaide: Australian Medicines Handbook; 2021 [cited 2021]. Available from: <https://childrens-amh-net-au.pklibresources.health.wa.gov.au/>.
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Related CAHS internal policies, procedures and guidelines
CAMHS Medication Safety Toolbox
CAMHS Physical Health Psychotropic Medication Monitoring Handbook
Psychotropic Medication Monitoring or Adverse Physical Health Effects
CAMHS Policies, Procedures, Guidelines
CAMHS Off-Label Prescribing Guide for CAMHS Clinicians
Useful resources (including related forms)
WCHN Monitoring Guidelines
Metabolic Syndrome in Children
Bridging the gap from evidence to policy and practice: reducing the progression to metabolic syndrome for children and adolescents on antipsychotic medication.

This document can be made available in alternative formats on request.

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